
RL Project Proposal

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1 Motivation

In contextual reinforcement learning, it is often assumed that training should progress from easy to difficult tasks. However, it is unclear whether such a simple-to-difficult design is always beneficial, especially when tasks differ in more than one aspect. At the same time, recent work has increasingly shifted away from manual, static curricula toward automatic curriculum learning. This motivates the question of how difficulty should actually be balanced in automatic task or context curricula.

2 Related Topics

This project is related to contextual MDPs, generalization in reinforcement learning, and automatic curriculum learning. It also connects to PPO as a standard policy optimization method and to benchmark settings such as CARL, where environment variations are represented through explicit contexts.

3 Idea

The main idea is to study how automatic context curricula should balance task difficulty during training. Instead of assuming that a fixed easy-to-hard schedule is optimal, we want to compare different ways of choosing contexts automatically based on signals such as performance, prediction error, or uncertainty. Finally, we want to investigate what should happen when certain difficult contexts are too hard to solve early in training, for example whether they should be postponed, down-weighted, or revisited later.

$$\pi_\theta \in \Pi, \quad \pi_\theta : \mathcal{S} \times \mathcal{C} \mapsto \mathcal{A} \tag{1}$$

4 Experiments

We plan to compare several automatic curriculum strategies and evaluate how they affect training and generalization across contextual benchmark environments.

Environments & Metrics We will use multiple CARL environments, starting with at least one simpler and one more challenging task. Candidate environments include contextual CartPole and contextual LunarLander. We plan to measure episodic return, sample efficiency, robustness across seeds, and generalization to unseen contexts. In addition, we want to evaluate curriculum-specific signals such as error-based and uncertainty-based difficulty estimates.

Experimental Scope The main comparison will include at least three curriculum strategies: uniform sampling, an error-based curriculum, and an uncertainty-aware curriculum. We will also compare explicit-context and hidden-context settings. Another planned comparison is how different methods

handle contexts that are initially too hard, for example by delaying them or by sampling them less frequently early on.

5 Timeline

We expect the project to proceed in the following stages:

- Research: 1 week
- Implementation: 2–3 weeks
- Experiments: 1 week
- Analysis: 4–6 days
- Making the final presentation: 1 week